

Education

- Nov 2021 – **International Institute of Information Technology Hyderabad.**
July 2026 B.TECH IN COMPUTER SCIENCE AND MS (BY RESEARCH) IN COMPUTATIONAL LINGUISTICS
- CGPA: 8.94/10.00 (Total); 9.28/10.00 (MS)
 - Dean's Academic List: 2021–22 (Top 5%), 2022–23 (Top 10%), 2023–24 (Top 15%)
 - Dean's Research List: 2024–2025 (Top 5%)
 - MS Thesis: "When Models Miss the Plot: Evaluating Reliability in Multimodal Reasoning over Charts, Grids and Maps", advised by Prof. Manish Shrivastava

Publications

- 2026 **RUBICON: Do Language Models Know When They've Crossed the Point of No Return?**
S. Mukhopadhyay*, A. Qidwai*, J. Sen, M. Shrivastava
EMNLP 2026  Preprint soon
- 2026 **PrivacyBench: A Conversational Benchmark for Evaluating Privacy in Personalized AI.**
S. Mukhopadhyay, S. Reddy, S. Muthukumar, J. An, P. Kumaraguru
WCCI (IJCNN) 2026, **Nominated for Best Paper**  Paper
- 2025 **InterChart: Benchmarking Visual Reasoning Across Decomposed Charts.**
A. Iyengar*, S. Mukhopadhyay*, A. Qidwai*, S. Singh, D. Roth, V. Gupta
AAACL 2025  Website
- 2025 **MapIQ: Evaluating Multimodal Large Language Models for Map QA.**
V. Srivastava, F. Lei, S. Mukhopadhyay, V. Gupta, R. Maciejewski
COLM 2025  Paper
- 2025 **PRAISE: Enhancing Product Descriptions with LLM-Driven Structured Insights.**
A. Qidwai*, S. Mukhopadhyay*, P. Khatiwada*, D. Roth, V. Gupta
ACL 2025 (System Demonstrations)  Website
- 2025 **MAPWise: Evaluating Vision-Language Models for Advanced Map Queries.**
S. Mukhopadhyay, A. Rajgaria, P. Khatiwada, M. Shrivastava, D. Roth, V. Gupta
NAACL 2025 (Oral), **Nominated for Outstanding Paper**  Website
- 2024 **Unraveling the Truth: Do VLMs really Understand Charts?**
S. Mukhopadhyay*, A. Qidwai*, A. Garimella, P. Ramu, V. Gupta, D. Roth
EMNLP 2024  Website

Research Experience

- May 2026 – **Anthropic Fellows / MATS Fellowship.**
Present Advisors: Rowan Wang (Anthropic), Christine Ye (Anthropic)
- Studying the science of overoptimization and reward hacking in LLM-judged Reinforcement Learning (RL) to understand when policies learn to exploit the judge instead of learning the actual task.
 - Found evidence of universal overoptimization, where a policy learns hacks that fool many different judges at once, undermining the use of held-out judges to track oversight progress.
 - Designed and created around 50 verifiable and non-verifiable RL environments, including chess, Go, research ideas, SQL generation, and creative writing, to find where models naturally learn to hack their judges during training.
 - Isolated causes of reward hacking using judge-capability ladders and judge-visibility ablations, understanding the effects of judge capability and robustness.
 - Built Litmus, a transcript viewer and hypothesis checker for monitoring, at a high level, how model behavior changes over the course of a large RL run.

Feb 2026 – **Supervised Program for Alignment Research (SPAR).**

May 2026 Advisor: Niels Warncke (Center on Long-Term Risk), Chad DeChant (New York University)

- **Harmful effects of episodic memory in LLM agents**
 - Built a framework that automatically generates long-horizon scenarios (up to 50 turns) that explicitly require the agent to have reliable long-term memory.
 - Analyzed whether having memory changes the agent's behavior in harmful ways, such as becoming better at sabotage, sandbagging, or lying to achieve its goals.
- **Unwanted generalization from fine-tuning**
 - Studied cases where narrow fine-tuning generalizes badly (as in emergent misalignment), e.g. whether a model fine-tuned on false facts from synthetic documents starts lying in other, unrelated situations as well.
 - Tested localized fine-tuning, where we selectively update only a chosen subset of a model's layers, in order to prevent this kind of unwanted generalization behavior.

May 2025 – **Whitebox AI Safety Fellowship.**

June 2025 Mentor: Kyle Reynoso (SASH)

- Performed interpretability research using model diffing with crosscoders to localize the emergence of misalignment in Large Language Models (LLMs).
- Project was ranked Top 3 in the cohort.

Aug 2023 – **Language Technologies Research Center (LTRC), IIIT Hyderabad.**

June 2026 Advisor: Prof. Manish Shrivastava (IIIT Hyderabad)

- Studied robustness in multimodal models to understand the gaps in their understanding and perception, and what causes robustness to break.
- Created a benchmark that places models in impossible situations, where no valid answer exists, to elicit whether they stay honest or fabricate an answer.
- Conducted comparative analysis of diverse LLMs to map representational and behavioral similarities and identify exactly where they diverge.
- Studied phase changes during pre-training and later during fine-tuning, approaching developmental interpretability through a behavioral lens.

Jan 2025 – **Precog Research Group, IIIT Hyderabad.**

May 2026 Advisor: Prof. Ponnurangam Kumaraguru (IIIT Hyderabad)

Collaborators: Prof. Jisun An (Indiana University), Prof. Saurav Prakash (IIT-M), Prof. Sheetal Kalyani (IIT-M)

- Evaluated privacy risks in personalized agents to detect secret leakage through the lens of Contextual Integrity frameworks.
- Investigated fairness constraints in Federated Learning (FL) setups to mitigate algorithmic bias across heterogeneous clients.
- Optimized the training of Binarized Neural Networks (BNNs) to enable efficient learning and adaptation on resource-constrained devices.

Jan 2024 – **Cognitive Computation Group (UPenn) / CORAL Lab (ASU).**

Aug 2025 Advisors: Prof. Dan Roth (UPenn), Prof. Vivek Gupta (ASU)

Collaborators: Dr. Aparna Garimella (Adobe), Pritika Ramu (Adobe), Prof. Ross Maciejewski (ASU)

- Investigated the robustness and consistency of Vision-Language Models (VLMs) in interpreting complex visual information, specifically charts, graphs, and geospatial maps.
- Benchmarked the robustness of VLM reasoning capabilities on different kinds of charts, including multiple charts, identifying failure points.
- Developed "MAPWise" and "MapIQ" benchmarks to evaluate multimodal models on advanced geospatial queries, counterfactual reasoning, and map-reading literacy.

Jan 2024 – **Software Engineering Research Center (SERC), IIIT Hyderabad.**

May 2024 Advisor: Prof. Karthik Vaidhyathan

- Investigated automated code summarization techniques to extract functional natural language descriptions from code snippets, benchmarking on the APPS dataset.
- Explored the integration of Abstract Syntax Trees (ASTs) to enrich model inputs with hierarchical structural context, aiming to enhance the semantic accuracy of generated summaries.

Industry Experience

Aug 2025 – **Interpret.**

Apr 2026 Role: Machine Learning Research Intern

- Created a semantic cache and related retrieval optimizations, cutting the pipeline's costs by up to 50% while improving recall.
- Built evaluations to find the limitations of our agentic system, then shipped fixes for the weakest areas.
- Created a pipeline for assigning more contextual names to taxonomy nodes.

May 2025 – **Google.**

Aug 2025 Role: Software Engineering Intern

Team: Google Cloud Storage (GCS) Orion

- Created a topology visualization tool for GCS Orion, enabling engineers to isolate and debug critical network issues across 100,000+ nodes.
- Received Pre-Placement Offer (PPO) for Full-Time Software Engineer role.

May 2024 – **Google.**

July 2024 Role: Software Engineering Intern

Team: Android PDF

- Designed and implemented the core image selection pipeline for the Android PDF viewer.
- Received Return Internship Offer for Summer 2025 based on exceptional performance.

May 2023 – **Google.**

July 2023 Role: STEP Intern

Team: Google Cloud Storage (GCS) Metering

- Built integrity checks for a high-throughput pipeline processing petabytes of Cloud usage data, ensuring billing accuracy for enterprise customers.
- Converted to standard Software Engineering Intern (Return Offer) for Summer 2024.

Honors and Awards

- **Heidelberg Laureate Forum (2026):** Selected as one of 100 young researchers in computer science worldwide to attend the 13th HLF in Heidelberg, Germany.
- **India AI Fellowship (2025):** Fellowship and aid to conduct research on understanding representational similarities across different LLMs.
- **Amazon ML Challenge (2024):** Secured **13th Rank** nationally among ~75,000 participants (Top 0.02%).
- **Google Generation Scholarship (2024):** Awarded for academic excellence and leadership in technology.
- **JPMorgan Chase Quant Research Mentorship (2024):** Selected for the specialized quantitative research track.
- **D. E. Shaw Ascend Educare (2022):** Selected as a Fellow for the prestigious mentorship cohort as one of the youngest participants.

Teaching and Leadership

Teaching Assistantships

2023 – 2025 **Teaching Assistant, IIIT Hyderabad.**

Head TA for Introduction to NLP (Spring 2024, 2025); TA for Language and Society (Monsoon 2024) and Algorithm Analysis & Design (Monsoon 2023).

Volunteering

2023 – 2024 **International Linguistics Olympiad (IOL).**

Team Leader & Mentor for the Indian National Team at the global finals (Bansko, Bulgaria); Lead Student Coordinator for the Panini Linguistics Olympiad, running the national selection process for thousands of participants.

2023 – 2024 **Club Coordinator, Open Source Developers' Group (OSDG).**

Organized an intra-college beginner hackathon with 150+ participants; conducted workshops on Linux, open source contributions, and Google Summer of Code.